

Project Title
Digital Library Search System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
REVU ROHITH VARMA (2520030070)
THOGARPALLY ADVIK (2520039623)

Team Number: 2
Section: 3

Under the guidance of

Dr. S. Vinay Kumar,
Associate Professor, Department of Computer Science and Engineering.



Hyderabad-500090, Telangana, India.

ABSTRACT

The Digital Library Search and Advanced Text Analytics System is a Java-based application that processes a local digital-library corpus through multiple DSA-3 algorithmic techniques. It provides keyword and pattern search, fuzzy query processing, document similarity, citation-flow analysis, scheduling and optimization demonstrations, and large-number validation without using a database. The system emphasizes efficient processing of real textual data through file-based corpus management.

Existing digital-library and text-search systems commonly depend on databases, indexing frameworks, or advanced search engines. The uniqueness of this project is its algorithm-centric and database-independent design, where core DSA-3 concepts are implemented and demonstrated directly on project corpus data. The system integrates string algorithms, dynamic programming, network flow, approximation, randomized and parallel techniques.

Key features include Naïve and KMP pattern matching, Z-function and Rabin-Karp concepts, suffix-based similarity, edit distance and sequence alignment, flow-based citation analysis, NP-hard scheduling with approximation, randomized hashing and primality testing, and parallel prefix-sum with performance analysis. The system is developed in Java using local text files as its corpus.

Signature of the Guide